Design, simulate and measure a series of Mach-Zehnder Interferometers

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1. Introduction

A Mach-Zehnder Interferometer (MZI) is a device that exploits the interference between optical fields by changing the relative phase shift between the two arms of the design. It is a fundamental building block in silicon photonic design that can be used in designing components such as switches, splitters and modulators.

In this work we will design, simulate and measure our imbalanced MZI designs to extract the free spectral range (FSR) which will then allow us to determine the group index (n_g) of the waveguide. Our simulations and measurement results will validate our theory and model.

2. Theory

In the context of this course, we can look at a MZI as consisting of a Y-branch splitter which divides your input light into 2 arms, followed by a Y-branch combiner which then combines your optical fields. In between these 2 components, we have waveguides that determine our respective path lengths. Consider this outline of an MZI with the description

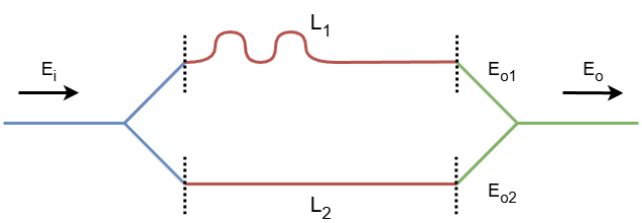

Figure 1. Outline of an imbalanced Mach-Zehnder interferometer.

Suppose we have an input electric field $E = E_i e^{-i(\beta z - \omega t)}$, with the propagation constant $\beta = \frac{2\pi \cdot n_{eff}}{\lambda}$. This field is passed through the Y-branch splitter (blue) and has propagated its respective path length (red) to arrive at the Y-branch combiner (green), given that an ideal 50/50 splitter divides light equally, and the $I \propto |E|^2$ relationship, the resulting electric fields is

$$E_{o1} = \frac{E_i}{\sqrt{2}} e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} \tag{1}$$

$$E_{o2} = \frac{E_i}{\sqrt{2}} e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \tag{2}$$

Notes
- we are omitting the time dependent phase term ωt.
- α is real and represents the power attenuated per unit length.

The output of the Y-branch combiner is then the sum of the electric fields divided by $\sqrt{2}$.

$$E_o = \frac{1}{\sqrt{2}}(E_{o1} + E_{o2}) \tag{3}$$

which then leads us to this expression

$$E_o = \frac{E_i}{2}\left(e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2}\right) \tag{4}$$

with intensity being expressed as

$$I_o = \frac{I_i}{4}\left|e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2}\right|^2 \tag{5}$$

From this, we consider lossless propagation, with identical waveguides leading to equivalent propagation constants $\beta = \beta_1 = \beta_2$ and expression for intensity as

$$I_o = \frac{I_i}{2}[1 + \cos(\beta \Delta L)] \tag{6}$$

Equation (6) represents the transfer function for an **imbalanced interferometer**. In contrast, a balanced interferometer has $\Delta L = 0$, whereby constructive and destructive interference must instead arise from variations in the propagation constant (Δβ).

With regards to the phase term in Eq. (6), this suggests we will periodically vary the intensity with the relative phase. This periodic phase condition leads directly to the concept of the free spectral range, which describes the wavelength spacing between adjacent transmission maxima. We begin with

$$\Delta\delta = \delta_m - \delta_{m+1} = 2\pi \tag{7}$$

$$\Delta\delta = (\beta_m - \beta_{m+1}) \cdot \Delta L = 2\pi \tag{8}$$

Assuming that β varies linearly with λ, this allows us to approximate $\Delta\beta$ as

$$\Delta\beta = (\beta_m - \beta_{m+1}) \approx -\frac{d\beta}{d\lambda}\Delta\lambda \tag{9}$$

Combining Eq (8) and Eq (9)

$$\Delta\lambda \approx -\frac{2\pi}{\Delta L}\left(\frac{d\beta}{d\lambda}\right)^{-1} \tag{10}$$

Since $\beta = \frac{2\pi \cdot n(\lambda)}{\lambda}$, the derivative with respect to wavelength is

$$\frac{d\beta}{d\lambda} = \frac{2\pi}{\lambda}\frac{dn}{d\lambda} - \frac{2\pi n}{\lambda^2} = -\frac{2\pi}{\lambda^2}\left(n(\lambda) - \lambda\frac{dn(\lambda)}{d\lambda}\right) \quad (11)$$

From which Eq (10) and (11) can be combined to give

$$\Delta\lambda = \frac{\lambda^2}{\Delta L \cdot \left(n(\lambda) - \lambda\frac{dn(\lambda)}{d\lambda}\right)} \quad (12)$$

Thus, we see that constructive and destructive interference is also dependent on wavelength, from which were derived from the phase relationships. We define this parameter as the free spectral range FSR Rewriting Eq (12),

$$FSR = \frac{\lambda^2}{\Delta L \cdot n_g} \quad (13)$$

With the group index defined as

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda\frac{dn_{eff}(\lambda)}{d_\lambda} \quad (14)$$

Thus, by measuring the FSR, we will be able to determine the group index n_g.

3. Modelling and Simulations

We will be using a strip waveguide of width $500nm$ and height $220nm$ in this course. Our simulation parameters are centered at $1550nm$ and we will be just considering the first quasi-TE polarized light. The waveguide mode profile is presented below.

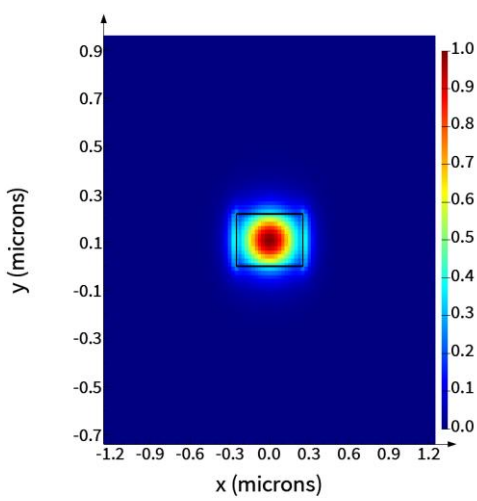

Figure 2. Waveguide mode profile showing electric field intensity of first quasi-TE mode.

Performing a wavelength sweep from 1500 to 1600 nm on our first quasi-TE mode, we find that the effective index decreases with wavelength while our group index increases.

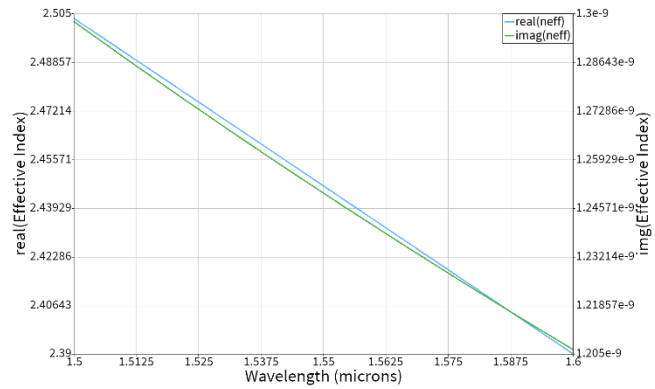

Figure 3. Effective Index vs Wavelength

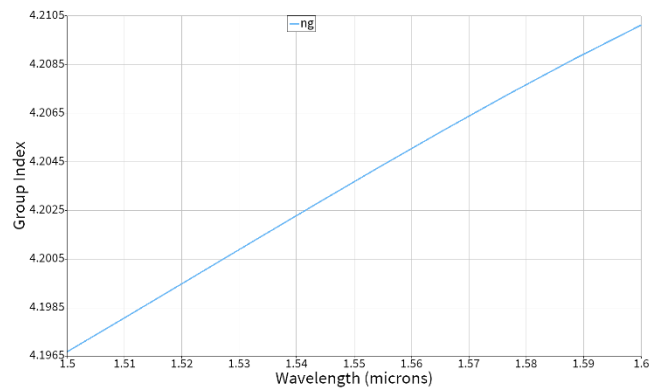

Figure 4. Group Index vs Wavelength

The compact model of this waveguide can then be calculated in **MATLAB** and expressed as

$$n_{eff}(\lambda) = 2.447 - 1.134(\lambda - 1.55) - 0.0368(\lambda - 1.55)^2 \quad (15)$$

where wavelength λ is in microns $[\mu m]$. This equation gives us sufficient information to model n_g as well.

4. Imbalanced Interferometer

We will be designing MZIs with varying path length differences, quantifying our expectations by both calculating the FSR using Eq. (13), and simulating the FSR from **INTERCONNECT**. Note that the simulated FSR values are approximate, as they are extracted from the FSR vs Wavelength plots.

For $1550nm$, and group index n_g = 4.204 we get

ΔL $[\mu m]$	calc. FSR $[nm]$	sim. FSR $[nm]$
80	7.143	7.00
100	5.715	5.72
130	4.396	4.40
140	4.082	4.07
220	2.598	2.58

The **EBeam library** in our INTERCONNECT simulations, where components in this library are modelled after real experimental data. The transfer functions of the Y-Branch and fibre grating coupler are shown below.

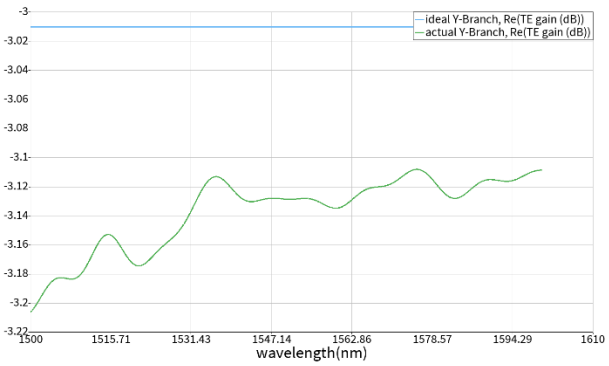

Figure 5. Transfer function of an ideal and typical Y-branch.

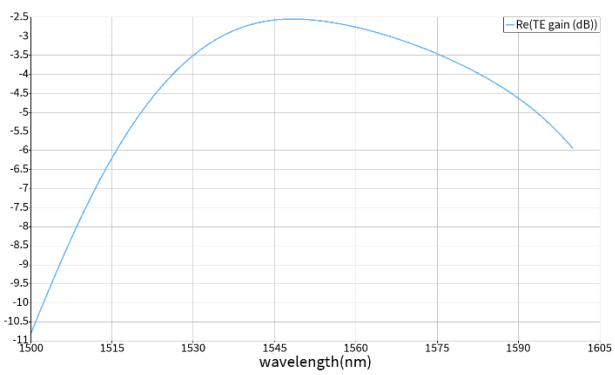

Figure 6. Transfer function of fibre grating coupler.

An example of the simulated MZI transfer function (or gain) is shown below. This configuration was for a path length difference of ΔL=140um, we note the impact the grating couplers have on the ideal transmission spectrum.

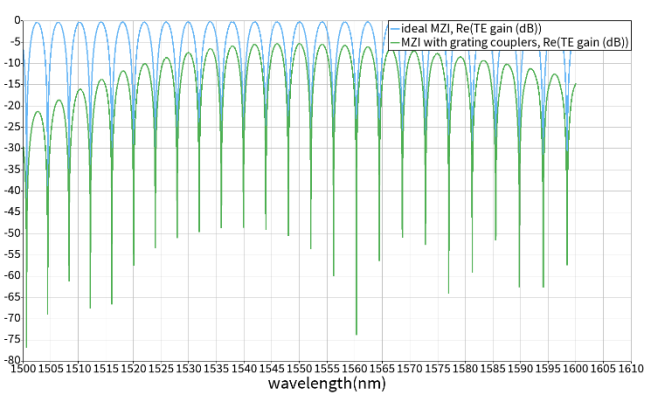

Figure 7. MZI transmission spectrum show with and without the grating couplers.

5. Design and Layout

KLayout was our tool of choice for the design portion of this project. Our allocated space was given as 605 x 410 μm, to which we managed 9 designs.

All designs were TE-polarised MZIs. These included 2 calibration loopback structures for de-embedding, a pair of MZIs that share the same path length with differing route geometries, and the rest of the MZIs with various path length differences.

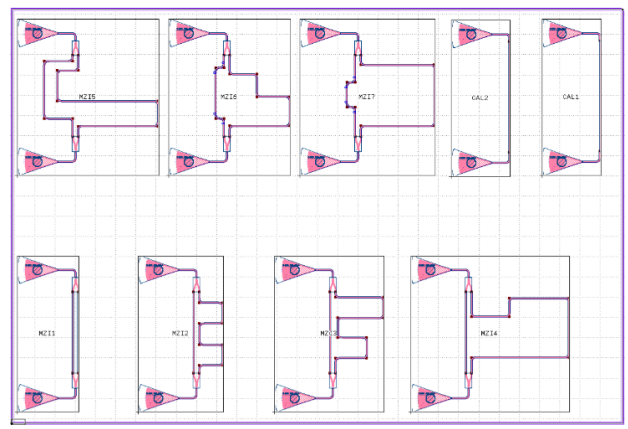

Figure 8. In KLayout, design space of 605 x 410 µm (purple border).

Label	designed ΔL [μm]
CAL1, CAL2, MZI1	0
MZI2	80
MZI6	100
MZI7	130
MZI3, MZI5	140
MZI4	220

Though we aim for a waveguide size of 500 x 220 nm, the manufacturing process will always have some dimensional variations. As such, we will perform a corner analysis, allowing us to evaluate the potential feature size variations. This technique effectively simulates edge cases to allow us to better manage expectations between simulations and practical measurements.

We will assume that the waveguide width may vary from 470 to 510 nm and waveguide height from 215.3 to 223.1 nm.

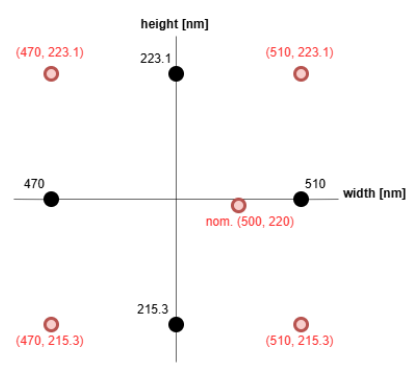

Figure 9. Visualising the target feature size variations. Corner analysis to be done on the 4 corners and nominal.

The **MODE** software was used to model the 5 waveguide size cases. This allowed the extraction of the waveguide properties, which were then imported into MATLAB for further processing and plotting.

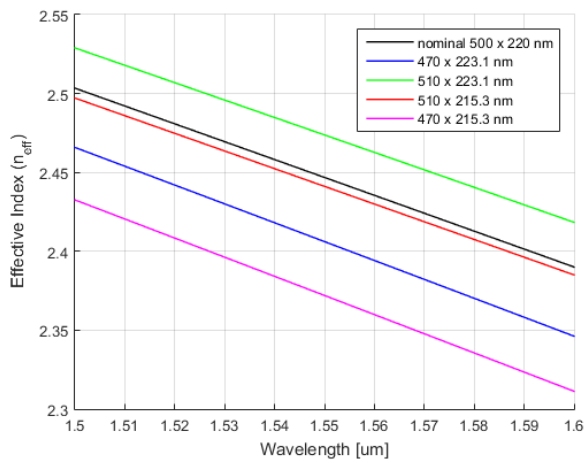

Figure 10. Simulated corner analysis for Effective Index vs Wavelength.

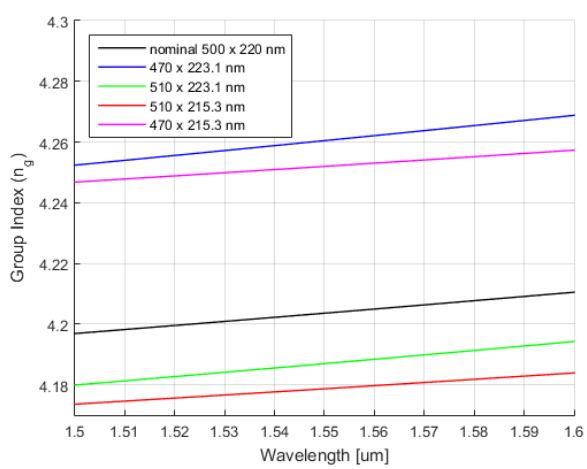

Figure 11. Simulated corner analysis of Group Index vs Wavelength.

With these feature size assumptions, we can identify the boundaries of the group index, namely **470 x 223.1 nm** will act as our upper simulated n_g bound while **510 x 215.3 nm**, our lower simulated n_g bound.

Given that the nominal group index lies within these boundaries, we expect the measured data to do so as well.

6. Fabrication Description

The photonic devices were fabricated using the NanoSOI MPW fabrication process by **Applied Nanotools Inc.** (http://www.appliednt.com/nanosoi; Edmonton, Canada) which is based on direct-write $100keV$ electron beam lithography technology. Silicon-on-insulator wafers of $200mm$ diameter, $220nm$ device thickness and $2\mu m$ buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25 x 25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H2SO4:H2O2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A $2.2\mu m$ oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300ºC. Reflectrometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

7. Measurement Description

To characterize the devices, a custom-built automated test setup [1, 5] with automated control software written in Python was used [2]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [3]. A 90º rotation was used to inject light into the TM grating couplers [3]. A polarization maintaining fibre array was used to couple light in/out of the chip [4].

8. Experimental Data and Discussion

In the interest of the report length, the measured data from **MZI4, MZI3** and **MZ5** will be the primary focus. Results from the remaining MZIs are summarised in tabulated form, where we note that the same analysis and procedure was performed.

The layout includes calibration loopback structures which are essentially the response of the 2 grating couplers. These structures let us de-embed the grating coupler weighting from the MZI response, resulting in a flatter transmission spectrum that more closely approaches the ideal MZI response.

The transmission spectrum of MZI4 is plotted below alongside one of the calibration structures.

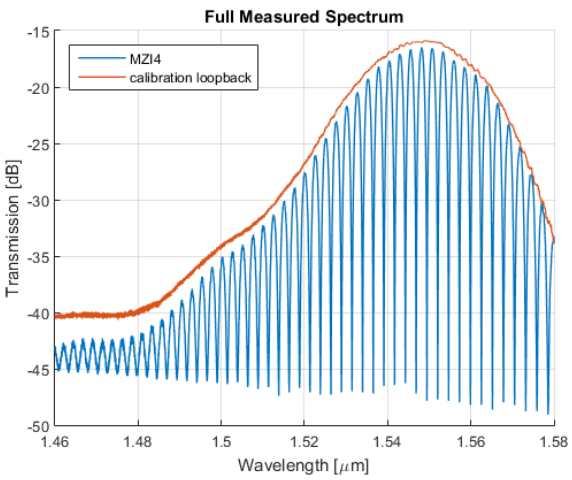

Figure 12. Measured spectrum of CAL2 (orange) and MZI4 (blue).

The shape of the calibration loopback response reflects the insertion loss introduced by the grating couplers.

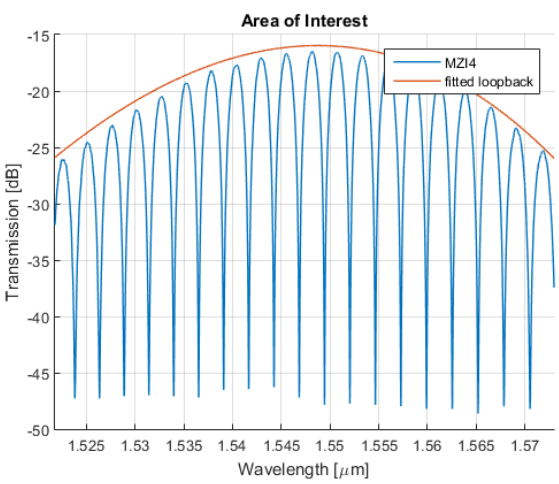

Figure 13. Truncated the wavelength range to be around 1550nm. The calibration loopback spectrum has also been curve fitted.

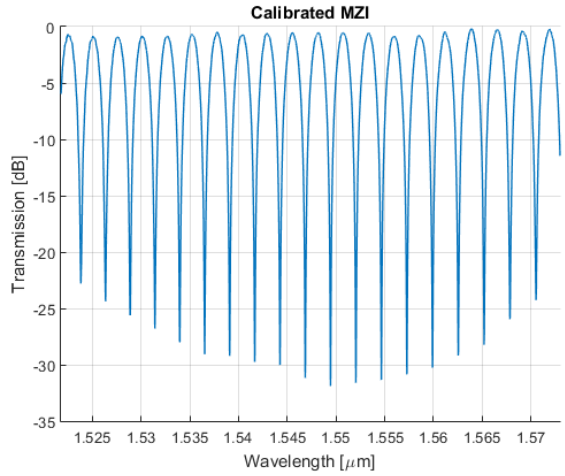

Figure 14. Calibrated MZI4 (de-embedded the calibration loopback).

From the de-embedded MZI4 transmission spectrum, we are then able to perform a curve fitting algorithm to extract our FSR and calculate our group index.

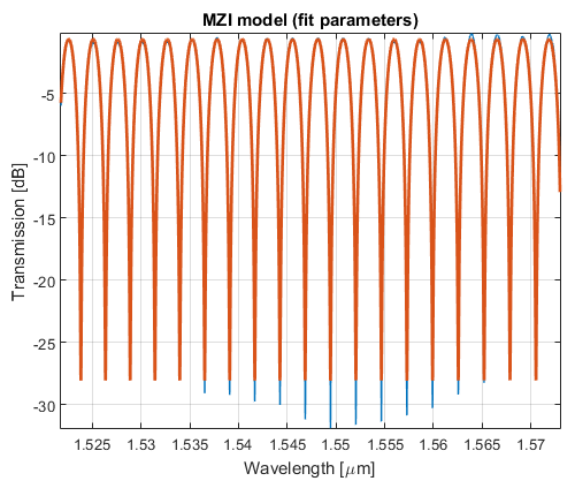

Figure 15. Calibrated MZI4 (blue), Curve fitted MZI4 (orange).

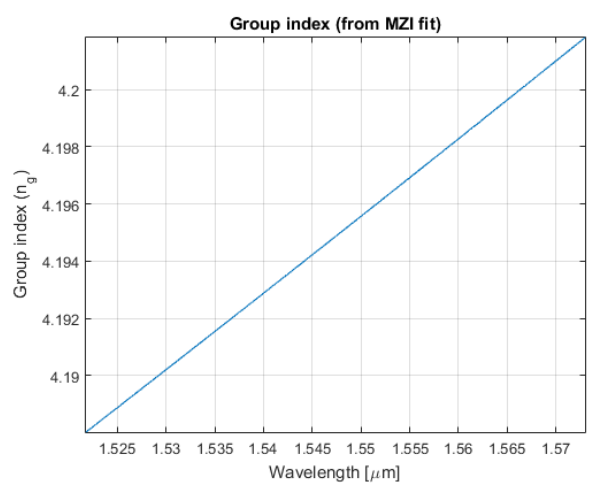

Figure 16. Experimentally calculated Group Index vs Wavelength.

At a wavelength of $1550nm$, we find that our measured FSR is $2.603nm$, corresponding to a calculated group index of 4.196.

Taking our percentage difference as

$$\frac{|x_{sim} - x_{meas}|}{{}^{1}/_{2} \cdot (x_{sim} + x_{meas})} \cdot 100 = \frac{|2.58 - 2.603|}{{}^{1}/_{2} \cdot (2.58 + 2.603)} \cdot 100 \approx 0.89\%$$

We find that our simulated and measured FSR show excellent agreement within 1%. Our group index value also shows good agreement as it lies within the boundaries established by our corner analysis.

Performing the same analysis with **MZI3** and **MZI5**, both share an identical path length difference of $\Delta L = 140\mu m$, but with different routing designs. For these MZIs, the simulated FSR was $4.07nm$.

Label	meas. FSR $[nm]$	FSR % diff.	n_g
MZI3	4.090	0.49	4.196
MZI5	4.095	0.61	4.191

The calculated group index lies within the bounds established by the corner analysis, confirming that both the FSR and group index are independent of route geometry. The plot below shows the resulting n_g against wavelength for MZI3, MZI4, and MZI5.

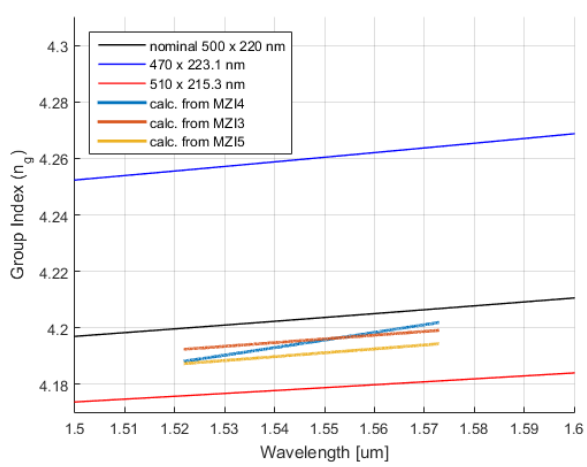

Figure 17. Comparing n_g extrema from the corner analysis and the experimenatlly calculated n_g. Group index vs Waveglength.

9. Conclusion

In conclusion, we have successfully designed, simulated and measured our Mach-Zehnder Interferometers. We find excellent agreement with our simulated and measured results, showing less than a 3% difference of the free spectral range (FSR) at the nominal wavelength of 1550nm. This let us extract the waveguide group index (n_g) which was shown to lie within the boundaries set up by our corner analysis which considered the potential waveguide dimensional variations common to the manufacturing process.

MZI3 and MZI5 (identical path length difference) showed nearly identical FSRs. This validates that the FSR is independent of routing geometry. Amongst all MZIs, the extracted group index was also consistent, with an average value of 4.195 at $1550nm$. This consistency is expected, as n_g is an intrinsic property of the waveguide rather than the interferometer layout.

Below is the tabulated rest of data (at $1550nm$). The FSR percentage difference is consistently less than 3%.

Label	ΔL $[\mu m]$	sim. FSR $[nm]$	meas. FSR $[nm]$	FSR % diff.	n_g
CAL1	0	n/a	n/a	n/a	n/a
CAL2	0	n/a	n/a	n/a	n/a
MZI1	0	n/a	n/a	n/a	n/a
MZI2	80	7.00	7.147	2.08	4.202
MZI3	140	4.07	4.090	0.49	4.196
MZI4	220	2.58	2.603	0.89	4.196
MZI5	140	4.07	4.095	0.61	4.191
MZI6	100	5.72	5.729	0.16	4.194
MZI7	130	4.40	4.410	0.23	4.191

10. Acknowledgements

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11. References

[1] Lukas Chrostowski, Michael Hochberg, chapter 12 in "Silicon Photonics Design: From Devices to Systems", Cambridge University Press, 2015

[2] http://siepic.ubc.ca/probestation, using Python code developed by Michael Caverley.

[3] Yun Wang, Xu Wang, Jonas Flueckiger, Han Yun, Wei Shi, Richard Bojko, Nicolas A. F. Jaeger, Lukas Chrostowski, "Focusing sub-wavelength grating couplers with low back reflections for rapid prototyping of silicon photonic circuits", Optics Express Vol. 22, Issue 17, pp. 20652-20662 (2014) doi: 10.1364/OE.22.020652

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[5] http://mapleleafphotonics.com, Maple Leaf Photonics, Seattle WA, USA.